1.26 Exponential Growth and Decay

Worked Solutions

1 a)

$y = 1 + 1 = 2$.

1 b)

$y = 20 - 1 = 19$.

1 c)

$y = 10 + 1 = 11$.

1 d)

$y = 5 - 1 = 4$.

1 e)

$y = 21 + 5 = 26$.

1 f)

$y = 3 - 5 = -2$.

2 a)

$P = \frac{100}{1+1} = 50$.

2 b)

$P = \frac{18}{1+1} = 9$.

3 a)

$P = 250$.

3 b)

$P = 250e^{0.15\times 20} = 5021$ (whole number as bacteria are discretely counted)

3 c)

$1000 = 250e^{0.15t} \Rightarrow 4 = e^{0.15t} \Rightarrow \ln 4 = 0.15t \Rightarrow \frac{1}{0.15}\ln 4 = t$ so $t = 9.24$ hours (3 SF)

4 a)

$6 = 3 \times 2^t \Rightarrow 2 = 2^t \Rightarrow t = 1$ hour.

4 b)

$8 = 4 \times e^{1.5t} \Rightarrow 2 = e^{1.5t} \Rightarrow 1.5t = \ln 2$
$\Rightarrow t = \frac{2}{3}\ln 2$ hours.

4 c)

$300 = 150 \times 1.6^t \Rightarrow 2 = 1.6^t$
$\Rightarrow t = \log_{1.6} 2$ hours.

4 d)

$48 = 20 + 4e^{0.1t} \Rightarrow 7 = e^{0.1t} \Rightarrow 0.1t = \ln 7$
$\Rightarrow t = 10 \ln 7$ hours.

5 a)

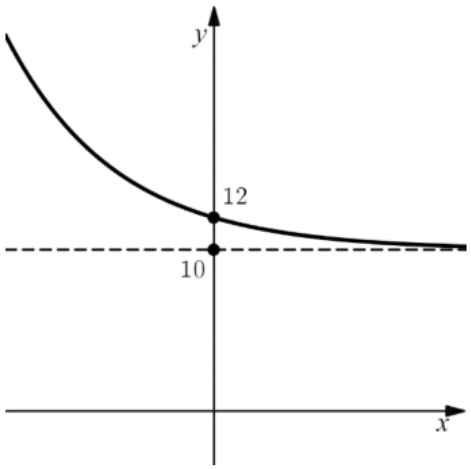
Initially $y = 1000$ grams.

5 b)

$\frac{1}{2} = 2^{-0.00142t} \Rightarrow \log_2 \frac{1}{2} = -0.00142t$
$\Rightarrow t = \frac{1}{0.00142} = 704.22535211\ldots \approx 704$ million.

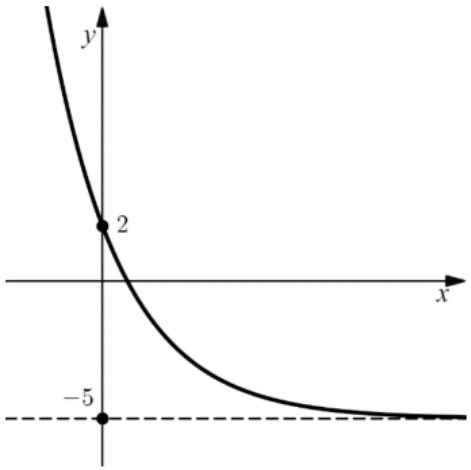
6 a)

$y = 10 + 2 = 12$, as $x \to \infty, y \to 10$



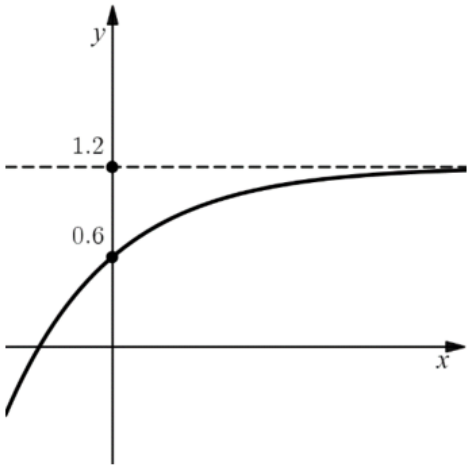
6 b)

$y = -7 + 7 = 2$, as $x \to \infty, y \to -5$



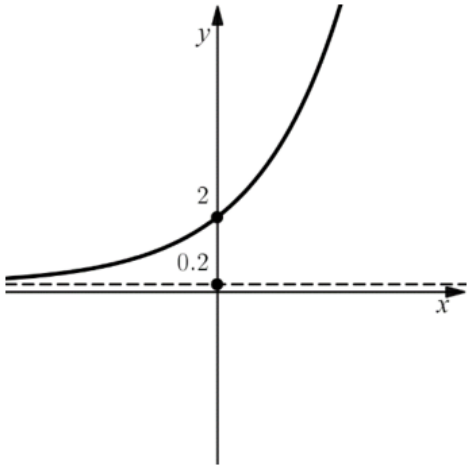
6 c)

$y = 1.2 - 0.6 = 0.6$, as $x \to \infty, y \to 1.2$



6 d)

$y = 0.2 + 1.8 = 2$, as $x \to \infty, y \to \infty$



7 a)

98°C.

7 b)

$y = 22$.

7 c)

$A = 22, B = 98 - 22 = 76$.

7 d)

Change the value of B to be 78.

7 e)

$A = 18$ and $B = 79$.

8 a)

$h = \frac{50}{1+4} = 10$ cm.

8 b)

As $t \to \infty, y \to \frac{50}{1+0} = 50$ cm.

9 a)

$h = 130 - 80 = 50$ cm.

9 b)

$130 - 80 \times 1.45^{-1.5} = 84.2$ cm.

9 c)

when $h = 100$,
$100 = 130 - 80 \times 1.45^{-x}$
$-30 = -80 \times 1.45^{-x}$
$\frac{3}{8} = 1.45^{-x}$
$-x = \log_{1.45}(\frac{3}{8})$ so $x = 2.63973\ldots$ years
32 months.

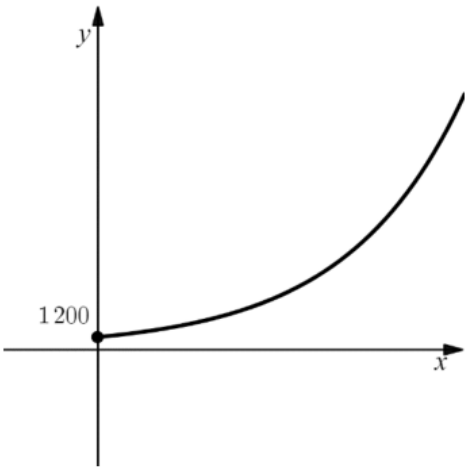
9 d)

As $x \to \infty, h \to 130$ so it doesn't go over a height of 1.3m.

10 a)

$y = 1200 \times 1.05^t$.

10 b)



10 c)

When $t = 4, y = 1200 \times 1.05^4 = £1458.61$.

10 d)

$2000 = 1200 \times 1.05^t \Rightarrow \frac{5}{3} = 1.05^t \Rightarrow t = \log_{1.05} \frac{5}{3} = 10.469848430\ldots$
So end of year 11.

11 a)

$\frac{225}{150} = 1.5$ so $\frac{150}{1.5} = £100$.

11 b)

$C = 100 \times 1.5^t$.

11 c)

$C = 100 \times 1.5^4 = £506.25$.

12 a)

$\frac{144}{180} = 0.8$ per 3 years which pass
$\frac{180}{0.8} = 225$ million active users initially.

12 b)

$P = 225 \times 0.8^{\frac{1}{3}t}$.

12 c)

$P = 225 \times 0.8^{\frac{1}{3} \times 10} = 106.9$ million active users.

13 a)

$\frac{3240}{1440} = 2.25$ per two months which pass
$\frac{1440}{2.25} = 640$ initially
$2.25^{0.5} = 1.5$ so $P = 640 \times 1.5^t$.

13 b)

640 is the initial population of insects, and 1.5 shows that the population 150% of its previous size as each month passes.

14 a)

$a = 12000$ and at $(5, 2500)$
$2500 = 12000e^{5k} \Rightarrow \frac{5}{24} = e^{5k}$
$\Rightarrow 5k = \ln \frac{5}{24} \Rightarrow k = \frac{1}{5} \ln \frac{5}{24}$
$P = 12000e^{\frac{1}{5} \ln \frac{5}{24} t}$.

14 b)

$P = 12000e^{2 \times \frac{1}{5} \ln \frac{5}{24}} = 6407.44299507 = 6407$ items.

14 c)

$500 = 12000e^{\frac{1}{5} \ln \frac{5}{24} t} \Rightarrow \frac{1}{24} = e^{\frac{1}{5} \ln \frac{5}{24} t}$
$\Rightarrow \frac{1}{24} = e^{\frac{1}{5} \ln \frac{5}{24} t} \Rightarrow \ln \frac{1}{24} = \frac{1}{5} \ln \frac{5}{24} t$
$\Rightarrow t = 10.1301210642 = 10.1$ months (3 SF)

14 d)

As $t \to \infty, P \to 0$ which doesn't allow for any changes in patterns such as ordering more stock.

15 a)

$C = Ae^{-kt}$
At $(0, 200) : A = 200$, at $(4, 100) : 100 = 200e^{-4k}$
$\Rightarrow \frac{1}{2} = e^{-4k} \Rightarrow \ln \frac{1}{2} = -4k \Rightarrow k = -\frac{1}{4} \ln \frac{1}{2} = \frac{1}{4} \ln 2$
$C = 200e^{-\frac{1}{4} \ln(2) t}$.

15 b)

$50 = 200e^{-\frac{1}{4} \ln(2) t} \Rightarrow \frac{1}{4} = e^{-\frac{1}{4} \ln(2) t}$
$\Rightarrow \ln \frac{1}{4} = -\frac{1}{4} \ln(2) t \Rightarrow -4 \ln \frac{1}{4} = \ln(2) t$
$\Rightarrow t = 8$ hours.

16 a)

$1000 = Ae^k$ and $14000 = Ae^{2k}$
Giving $\frac{1400}{1000} = e^k \Rightarrow e^k = 1.4$.

16 b)

$k = \ln 1.4$ and $A = \frac{1000}{1.4} = \frac{5000}{7}$, $P = \frac{5000}{7}e^{(\ln 1.4)n}$.

16 c)

$P = \frac{5000}{7}e^{4\ln 1.4} = 2744$ restaurants.

16 d)

$100000 = \frac{5000}{7}e^{\ln 1.4n} \Rightarrow 140 = e^{\ln 1.4n}$
$\Rightarrow \ln 140 = n \ln 1.4 \Rightarrow n = 14.6866\ldots$ so 15 years.

16 e)

As $t \to \infty, P \to \infty$ There can't be infinite restaurants.

17)

$0.01p_0 = p_0e^{-kt} \Rightarrow 0.01 = e^{-kt}$
$\Rightarrow \ln 0.01 = -kt \Rightarrow t = -\frac{1}{k}\ln 0.01 = \frac{1}{k}\ln 100$.

18 a)

At $(2, 500) : 500 = Ae^{-2k}$ At $(7, 88) : 88 = Ae^{-7k}$
$\frac{500}{88} = \frac{Ae^{-2k}}{Ae^{-7k}} \Rightarrow \frac{125}{22} = e^{5k} \Rightarrow \ln\frac{125}{22} = 5k \Rightarrow k = \frac{1}{5}\ln\frac{125}{22}$.

18 b)

Which gives $500 = Ae^{-\frac{2}{5}\ln\frac{125}{22}} \Rightarrow A = \frac{500}{e^{-\frac{2}{5}\ln\frac{125}{22}}}$
$\Rightarrow A = 1001.762885\ldots$ so $A = 1001.8$.

19 a)

$a + b = 85$ and $ae^{0.08\times 6} + be^{0.12\times 6} = 153$
$ae^{0.48} + (85 - a)e^{0.72} = 153$
$ae^{0.48} + 85e^{0.72} - ae^{0.72} = 153$
$ae^{0.48} - ae^{0.72} = 153 - 85e^{0.72}$
$a(e^{0.48} - e^{0.72}) = 153 - 85e^{0.72}$
$a = \frac{153-85e^{0.72}}{e^{0.48}-e^{0.72}} = 49.3358921682\ldots = 49.3$
$b = 35.7$.

19 b)

The approximate initial number of each type of fish.

19 c)

$49.3e^{0.08t} = 35.7e^{0.12t}$
$\frac{49.3}{35.7} = \frac{e^{0.12t}}{e^{0.08t}} = e^{0.04t} \Rightarrow 0.04t = \ln(\frac{49.3}{35.7})$
$\Rightarrow t = \frac{1}{0.04}\ln(\frac{49.3}{35.7}) = 8.06933480658$
After 8.07 months (3 SF)

19 d)

Both models predict infinite exponential growth but the fish populations will be limited by the size of the lake or may be affected by disease or predators.

20 a)

$T = 17 + ae^{-kt}$ At $(0, 65) : 65 = 17 + a$ so $a = 48$
$T = 17 + 48e^{-kt}$ At $(3, 35) : 35 = 17 + 48e^{-3k}$
So $18 = 48e^{-3k} \Rightarrow \frac{3}{8} = e^{-3k}$
$\Rightarrow \ln \frac{3}{8} = -3k \Rightarrow k = -\frac{1}{3} \ln \frac{3}{8} = \frac{1}{3} \ln(\frac{8}{3})$
$T = 17 + 48e^{-\frac{1}{3} \ln(\frac{8}{3})t}$.

20 b)

$T = 17 + 48e^{-\frac{5}{3} \ln(\frac{24}{7})} = 26.4°\text{C}$ (3 SF)

20 c)

As $t \to \infty, T \to 17°\text{C}$.

20 d)

The value "17" in the model would be reduced and the "48" would increase by the same amount.

20 e)

$y = 17 + 63e^{-\frac{1}{2} \ln(\frac{24}{9})t}$
$17 + 63e^{-\frac{1}{2} \ln(\frac{24}{9})t} = 17 + 48e^{-\frac{1}{3} \ln(\frac{24}{9})t}$
$\frac{63}{48} = \frac{e^{-\frac{1}{3} \ln(\frac{24}{9})t}}{e^{-\frac{1}{2} \ln(\frac{24}{9})t}} \Rightarrow \frac{63}{48} = e^{\frac{1}{6} \ln(\frac{24}{9})t}$
$\Rightarrow \ln \frac{63}{48} = \frac{1}{6} \ln(\frac{24}{9})t$
$\Rightarrow t = \frac{\ln \frac{63}{48}}{\frac{1}{6} \ln(\frac{24}{9})} = 1.66349269039\ldots$
$t = 1.66$ minutes (3SF)

21 a)

$\frac{dy}{dx} = -0.6e^{-0.3x}$
$\left.\frac{dy}{dx}\right|_{x=2} = -0.6e^{-0.6} = -0.329$ (3 SF)

21 b)

$\frac{dy}{dx} = 4 \times -0.2e^{0.2x}$
$\left.\frac{dy}{dx}\right|_{x=2} = 0.8e^{0.4} = 1.19$ (3 SF)

21 c)

$\frac{dy}{dx} = -48e^{-0.8x}$
$\left.\frac{dy}{dx}\right|_{x=2} = 48e^{-1.6} = 9.69$ (3 SF)

22)

$\frac{dP}{dt} = -\frac{140}{9} \ln(\frac{700}{57})e^{-\frac{1}{225} \ln(\frac{700}{57})t}$
$\left.\frac{dP}{dt}\right|_{x=200} = -\frac{140}{9} \ln(\frac{700}{57})e^{-\frac{200}{225} \ln(\frac{700}{57})}$
$= -4.19778030561$ thousand per year
≈ -4.2 thousand per year.

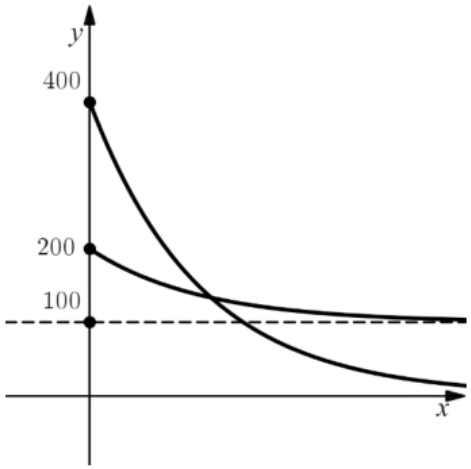
23 a)

As $t \to \infty, V \to 330$ml.

23 b)

$\frac{dV}{dt} = 33e^{-0.1t}$ so $\left.\frac{dV}{dt}\right|_{x=5} = 33e^{-0.5} = 20.0155117705$
20.0 ml/s (3 SF)

24 a)



24 b)

Amount of product A $\to 100$
Amount of product B $\to 0$.

24 c)

In both cases the models don't actually reach these values, and as the amount of each product is discrete the model only approximates at each point.

24 d)

A: $\frac{dP}{dt} = -100ke^{-kx}$
B: $\frac{dP}{dt} = -400ke^{-kx} = 4 \times -100ke^{-kx}$.

24 e)

$100(1 + e^{-2k}) = 400e^{-2k}$
$\Rightarrow 100 + 100e^{-2k} = 400e^{-2k}$
$\Rightarrow 100 = 300e^{-2k} \Rightarrow \frac{1}{3} = e^{-2k}$
$\Rightarrow \ln \frac{1}{3} = -2k \Rightarrow k = -\frac{1}{2} \ln \frac{1}{3} = \frac{1}{2} \ln 3$.

25 a)

$h = \frac{18}{1+3} = 4.5$ m.

25 b)

$h = \frac{18e^{\frac{1}{2}t}}{e^{\frac{1}{2}t}+3} \times \frac{e^{-\frac{1}{2}t}}{e^{-\frac{1}{2}t}} = \frac{18}{1+3e^{-\frac{1}{2}t}}$ as req.

25 c)

As $t \to \infty, h \to 18$ m.

25 d)

$2 = a - 14 \Rightarrow a = 16$.

25 e)

$\frac{dh}{dt} = 7e^{-\frac{1}{2}t}$
so $\left.\frac{dh}{dt}\right|_{t=5} = 7e^{-\frac{5}{2}} = 0.575$ metres per year (3 sf)

25 f)

As $t \to \infty, h \to 16$ m. So the first tree is expected to be taller.

26)

First model: $\frac{dm}{dt} = -5e^{-0.05t}$ so $\left.\frac{dm}{dt}\right|_{t=0} = -5$
Second model: $\frac{dm}{dt} = -90ke^{-kt}$
so $\left.\frac{dm}{dt}\right|_{t=0} = -90k$
$-90k > -5$ so $k > \frac{1}{18}$.